MA 17: HOW TO SOLVE IT
HANDOUT 2: ELEMENTARY NUMBER THEORY

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Extremal elements: from last time.

Problem 1. (Handout 1, Problem 10)
100 soccer teams participate in a tournament, with one match between every two teams. Assume that there are no draws, and that no team wins all 99 games. Prove that there exist three teams A, B, and C such that A beats B, B beats C, and C beats A.

Problem 2. (Putnam 2025, A1; Handout 1, Problem 11)
Let m_0 and n_0 be distinct positive integers. For every positive integer k, define m_k and n_k to be the relatively prime positive integers such that

$$\frac{m_k}{n_k} = \frac{2m_{k-1}+1}{2n_{k-1}+1}.$$

Prove that $2m_k + 1$ and $2n_k + 1$ are relatively prime for all but finitely many positive integers k.

Prime factorization and divisibility. Every integer $n > 1$ has a unique factorization

$$n = p_1^{a_1} \cdots p_r^{a_r},$$

up to reordering, where the p_i are distinct primes and $a_i \geq 1$. For a prime p and a positive integer n, write $v_p(n)$ for the exponent of p in n; in particular, $v_p(n) = 0$ if $p \nmid n$. Then

$$v_p(ab) = v_p(a) + v_p(b), \qquad v_p(\gcd(a,b)) = \min\{v_p(a), v_p(b)\}.$$

Problem 3.
Prove that the product of three consecutive positive integers is never a perfect square.

Problem 4.
Prove that

$$1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{N}$$

is not an integer for any integer $N \geq 2$.

Congruences. We write $a \equiv b \pmod{m}$ if $m \mid a - b$. Congruences may be added and multiplied; division by a modulo m is valid when $\gcd(a, m) = 1$.

Problem 5.
Prove that the equation

$$x^2 + 10y^2 = 3z^2$$

has no solution in positive integers x, y, z.

Fermat's little theorem. If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$. Equivalently, $a^p \equiv a \pmod{p}$ for every integer a.

Problem 6.
Prove that for every prime p, there is a positive integer n such that

$$p \mid 2^n + 3^n + 6^n - 1.$$

Multiplicative order and primitive roots. For a prime p and an integer a not divisible by p, the *order* of a modulo p is the least positive integer d such that $a^d \equiv 1 \pmod{p}$. It divides $p-1$, and $a^j \equiv 1 \pmod{p}$ if and only if $d \mid j$. A *primitive root modulo* p is an integer g of order $p-1$; its powers

$$1, g, g^2, \ldots, g^{p-2}$$

represent all nonzero residue classes modulo p. Every prime has a primitive root.

Problem 7. (Putnam 2021, A5, with a warm-up)
(a) Let p be a prime and j a nonnegative integer. Determine

$$\sum_{a=1}^{p-1} a^j \pmod{p}.$$

(b) Let A be the set of all integers n such that $1 \le n \le 2021$ and $\gcd(n, 2021) = 1$. For every nonnegative integer j, let

$$S(j) = \sum_{n \in A} n^j.$$

Determine all values of j such that $S(j)$ is a multiple of 2021.

Base representations. For any integer $b \ge 2$, every positive integer has a unique representation

$$n = a_r b^r + a_{r-1} b^{r-1} + \cdots + a_1 b + a_0, \qquad a_i \in \{0, 1, \ldots, b-1\}, \quad a_r \neq 0.$$

The cases $b = 2$ and $b = 10$ give binary and decimal representations.

Problem 8.
Let $k \ge 2$ be an integer. Find the sum of all integers n with $1 \le n \le 2^k - 1$ whose binary representation contains an even number of ones.

Problem 9. (Putnam 2020, A1)
How many positive integers N satisfy all of the following conditions?

- N is divisible by 2020.
- The decimal representation of N has at most 2020 digits.
- The decimal representation of N consists of a nonempty string of ones followed by a string of zeros.

Reminder PSet 2 to be released tomorrow, and due by the end of Oct 21 (on Canvas).